Senes Organs

Eye

Sense organs

- A specialized organ or structure such as the eye, ear, tongue, nose and skin, where sensory neurons are concentrated and functions as a receptors.
- These receptors are classified into:-

Enteroreceptor:

- Any receptor that responds to stimuli inside body.

Exteroreceptor:

Any receptor that respond to stimuli outside body.

Chemoreceotor:

- Any receptor that respond to chemical stimuli such as taste buds of the tongue and the olfactory epithelium of the nasal cavity which is responsible for smell.

Proprioceptor:

Which involved in detecting the body position, it includes:

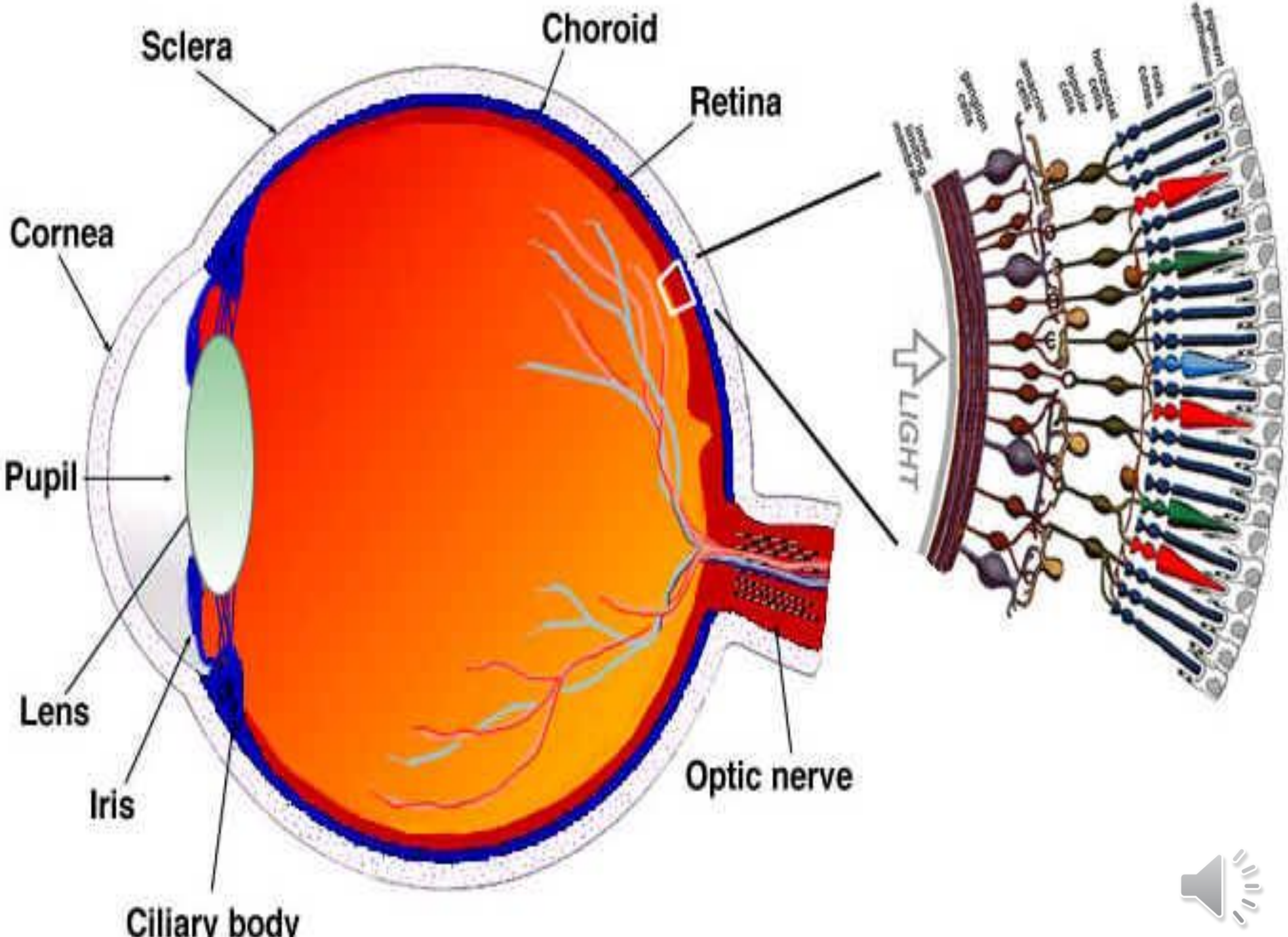
1. Muscle spindle.
2. Golgi tendon organ.

Thermoreceptor:

- Sensory receptor that respond to heat and cold

Eye

- The eye is the most efficient organ has evolved. It is not only reacting to various intensities and qualities of light but it is capable of distinguishing the form and the size of external objects.
- The eye is a 3 layered structures which is light tight except at one site, the cornea. Many tissues of the eye are heavily pigmented.


Sclera
Choroid
Retina
Cornea
Pupil
Lens
Iris
Ciliary body
Optic nerve
LIGHT

The eye is a 3 layered structures
1- Tunica fibrosa, 2- Tunica vasculosa
3- Tunica interna

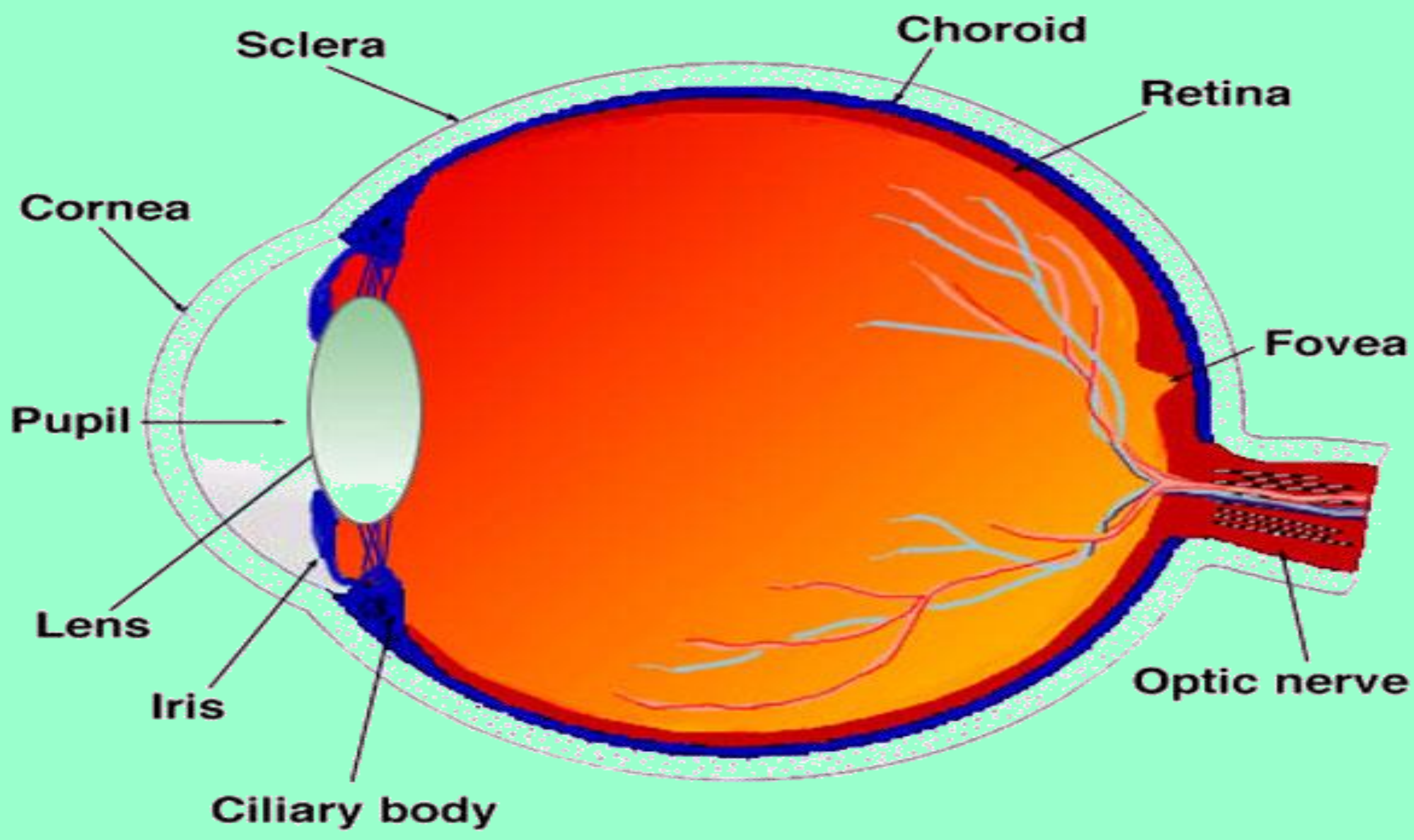


Fig. 6. Vertical sagittal section of the adult human eye.

I. Tunica fibrosa (External layer)

- It consists of cornea and sclera.
- **<u>Cornea</u>**
- It is the anterior one sixth of the eye.
- It is the primary site of refraction and has five layers (three of which are cellular).

- Corneal epithelium: is a non-keratinized, stratified squamous epithelium and exhibits constant repair. The epithelium is innervated by free nerve ending.
- Bowman's membrane: is a homogeneous and structure less under light microscope. However, electron micrographs show it consists of a network of randomly arranged collagen fibrils about 180 Ao in diameter, which may show a periodic banding.

It has a role in stability and strength of the cornea.

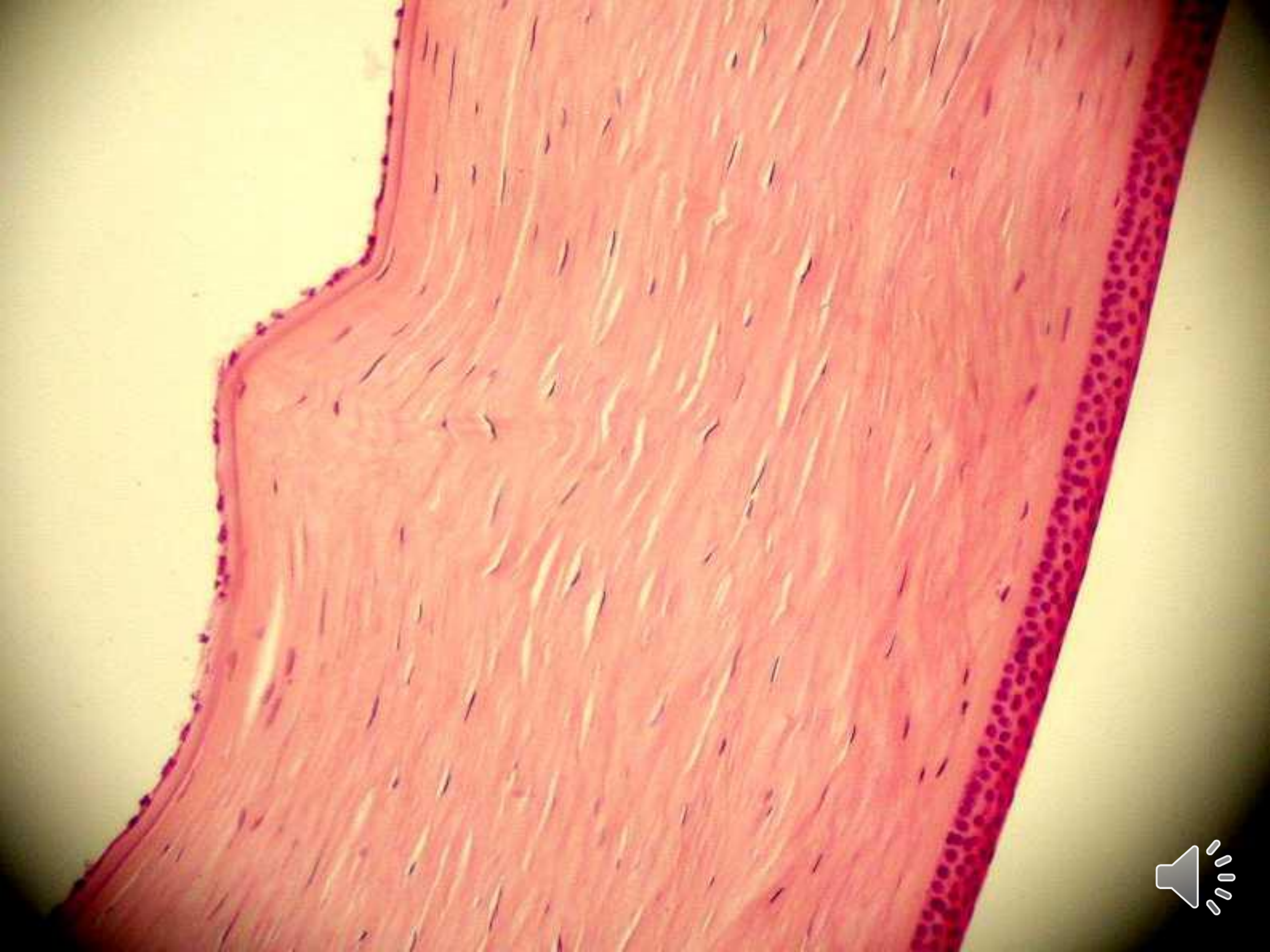
Bowman's membrane is absent in camel.

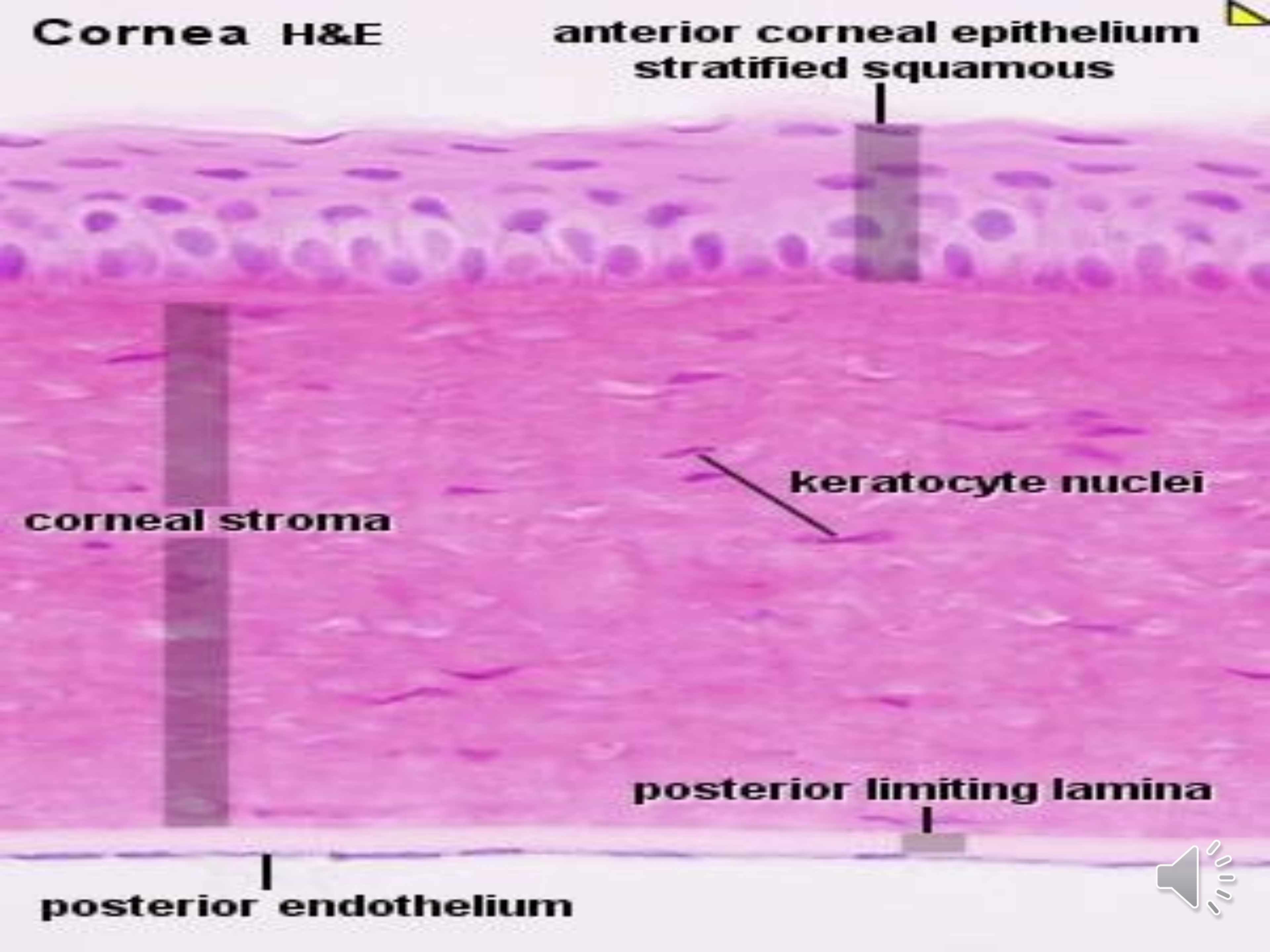
- Stroma (Substantia propria): It forms about 90% of the thickness of the cornea.

It is composed of many layers of parallel collagen bundles that cross at right angle to each other, this orientation and spacing of the collagen fibers is critical for the transparency of the cornea.

Fibroblast are flattened between the collagen called keratocytes.

- Descemet's membrane: it is a homogenous structure and composed of fine collagenous filaments organized in a three dimensional network.
- Endothelium: is a single layer of squamous or cuboidal epithelial cells that line the surface of the cornea exposed to the anterior chamber. The cells of the endothelium are critical for pumping fluid out of the stroma of the cornea, without this function, the stroma would swell with fluid and the optical properties would be affected.




Cornea H&E
anterior corneal epithelium
stratified squamous
keratocyte nuclei
corneal stroma
posterior limiting lamina
posterior endothelium

Sclera

- It forms the opaque posterior five-sixths of the protective outer tunic.
- It consists of flat collagenous bundles that run in various directions parallel to the surface. Between these bundles are network of elastic fibers, the cells of the sclera are flat elongated fibroblast.
- The tendons of the eye muscles are attached the outer surface of the sclera “episclera”, which in turn is connected with a dense layer of C.T. “The capsule of Tenon” by thin collagen fiber separated by clefts “space of tenon”. This arrangement Facilitate rotational movement of the eye ball in all directions.
- Between the sclera and the choroids is a layer of loose connective tissue with elastic fibers, melanocytes and fibroblasts. When these two tunics are separated part of this loose connective tissue adheres to the choroids and part to the sclera as suprachoroid lamina

Limbus (corneoscleral junction)

- It is an area of transition from the transparent cornea to the opaque sclera.
- In this area the collagenous bundles of the sclera continue directly into those of the cornea, where they become more precisely ordered, their striation loses its distinctness and the tissue becomes more homogeneous and transparent.
- This area is highly vascularized.
- **N.B** The cornea is devoid of blood vessels, receives its nutrition by diffusion from adjacent vessels and from the fluid of the anterior chamber.

- **Trabecular meshwork:-**
- It is located just peripheral to the end of Descemet's membrane. The trabecular strands which are continuous with Descemet's membrane and endothelium enclose spaces called "spaces of Fontana" that communicate with the anterior chamber.

- **Schlemm's canal:**

-It is an irregular endothelium. Lined channels which form canal in the stromal layer in the limbus region.

- It drain fluid from the anterior chamber of the eye any obstruction of this drainage causes glaucoma.

Choroid

It is divided into 5 layer as follows :-

1- Suprachoroidal layer

2- Vascular layer

3- Topetum Lucidum

4- Choriocapillary layer

5- Bruch’s membrane

II Tunica vasculosa (Middle layer)

- It consists of 3 parts: choroid, ciliary body and iris.

Choroid

- It forms the posterior part of the uvea.
- It is separated from the sclera by the perichoroidal space. The choroids is attached to the pigment epithelium layer of the retina, when retina is detached the pigment epithelium remains adherent to the choroids.

It is divided into 5 layer as follows.

1- Suprachoroidal layer:

- It consists of loosely arranged collagen fiber and some elastic fibers, fibroblasts and numerous melanocytes.

2- Vascular layer:-

This layer contains numerous large and medium. Sized arteries and veins. The space between the vessels are filled with loose connective tissue rich in melanocyles.

3- Tapetum Lucidum:-

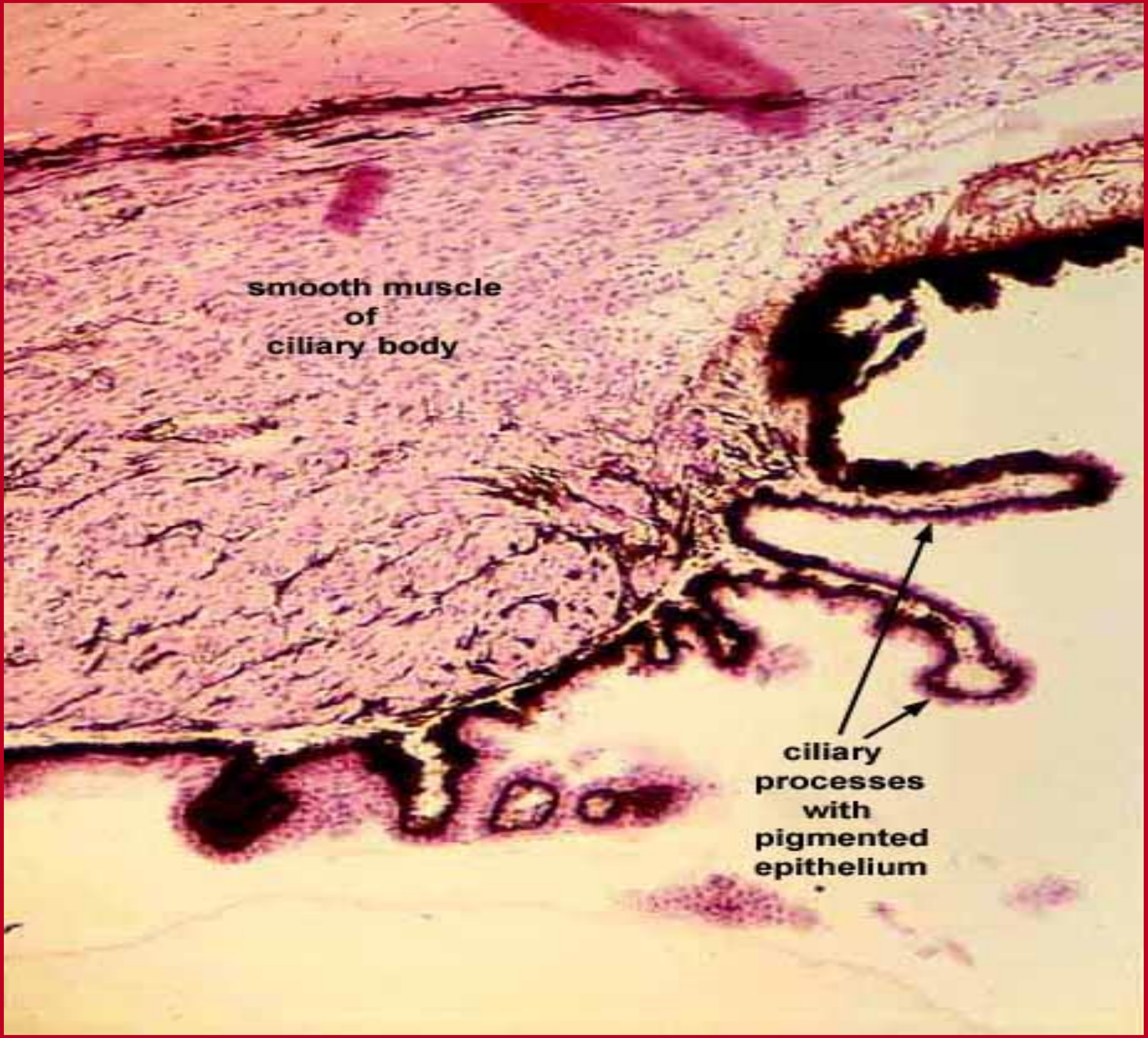
- It is a light-reflecting layer to increase light perception in poor illumination.
- In herbivores, the tapetum is fibrous consisting of intermingling collagen fibers with few fibroblast.
- In carnivores: it consists of layers of flat polygonal cells that appear brick-like (35 layers in cat and 15 layers in dog).
- The tapetal cells are packed with bundles of parallel small rods (modified melanosomes).
- The tapetum is absent in camel and swine

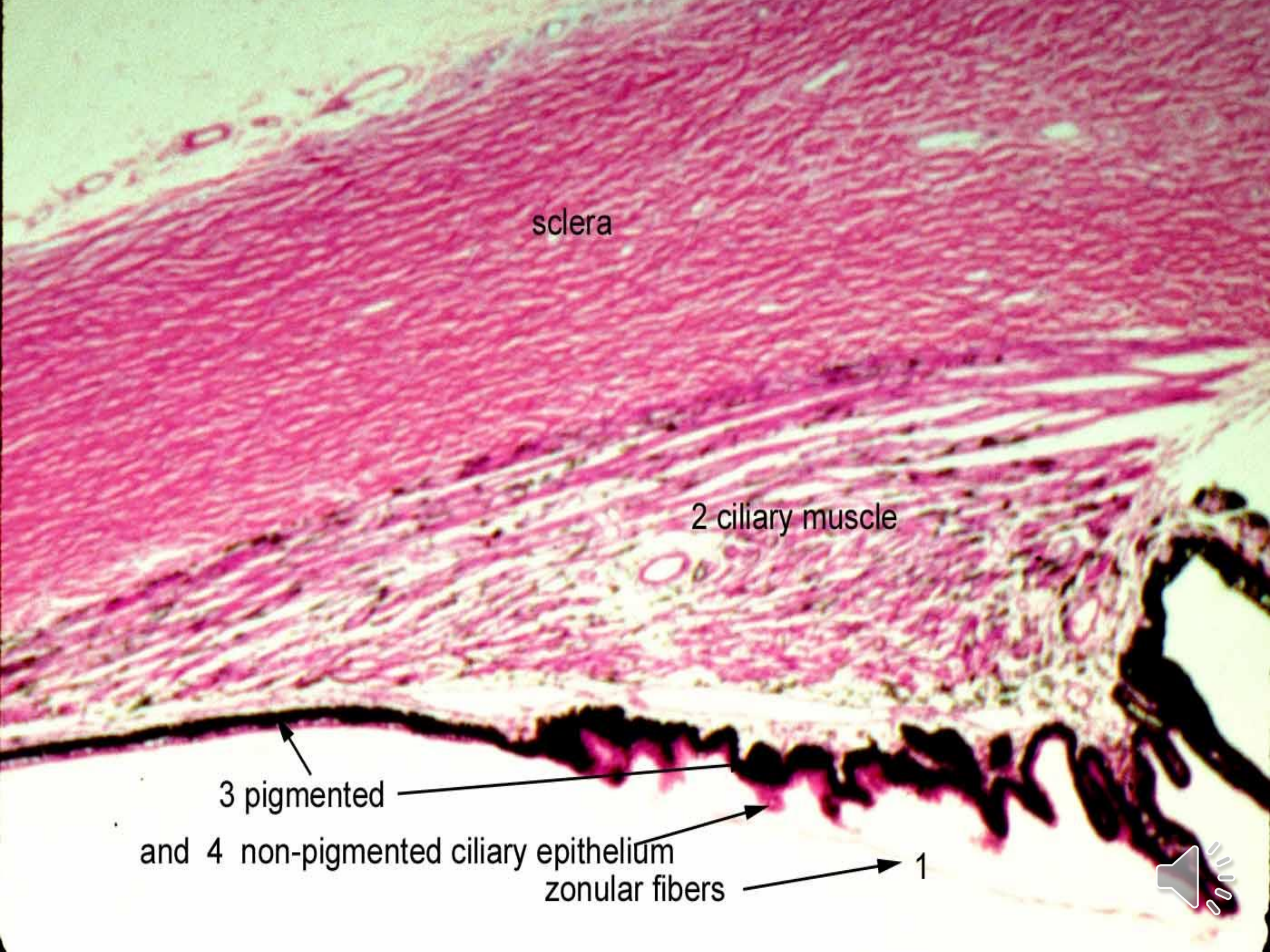
choroid

- **4- Choriocapillary layer:**
- - It is network of fenestrated capillaries.
- - It provides nutrition to the retina.
- **5- Bruch's membrane:**
- - It separates the Choriocapillary layer from the retina.
- - It consists of the basal lamina of the retinal pigment epithelium, a layer of collagen and elastic fibers and the basal lamina of the choriocapillary layer.

Ciliary body

- It is the direct rostral continuation of the choroids at the level of the lens.
- It is continuous with the iris rostrally.
- All layer of the choroids extend into ciliary body except tapetum and the choriocapillary layer.
- It consists of loose connective tissue rich in elastic fibers, BVS and melanocytes. It is covered by a bilayered epithelium.
- It has 2 specialized structures:-

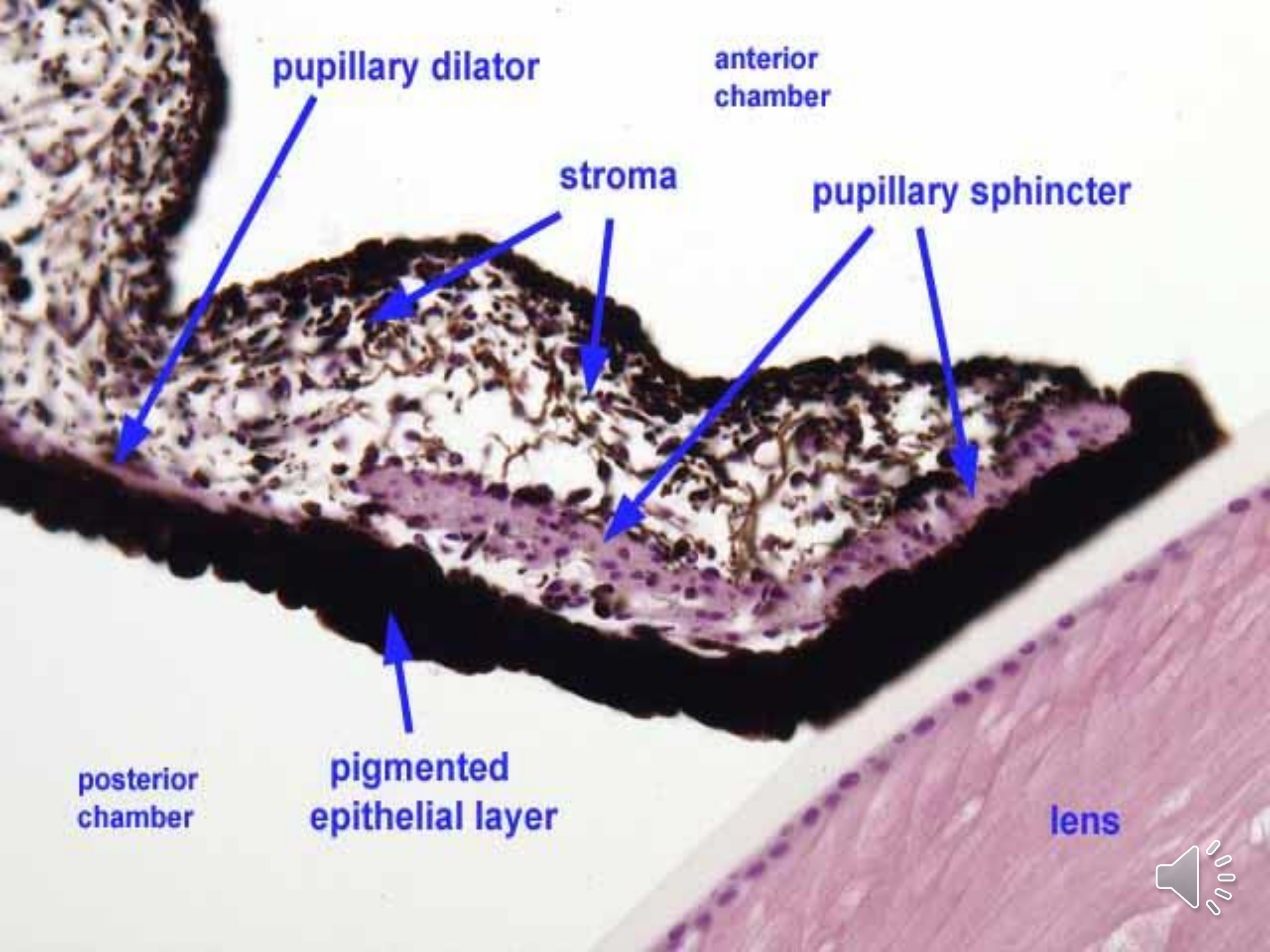

smooth muscle
of
ciliary body
ciliary
processes
with
pigmented
epithelium


sclera
2 ciliary muscle
3 pigmented
and 4 non-pigmented ciliary epithelium
zonular fibers
1

Iris

- The most anterior part of the uvea, the iris has a central aperture, the pupil.
- The iris divides the space between cornea and lens into the anterior chamber of the eye and the posterior chamber of the eye. The two communicate through the pupil.
- The anterior surface of the iris is formed of discontinuous layer of pigment cells and fibroblast.

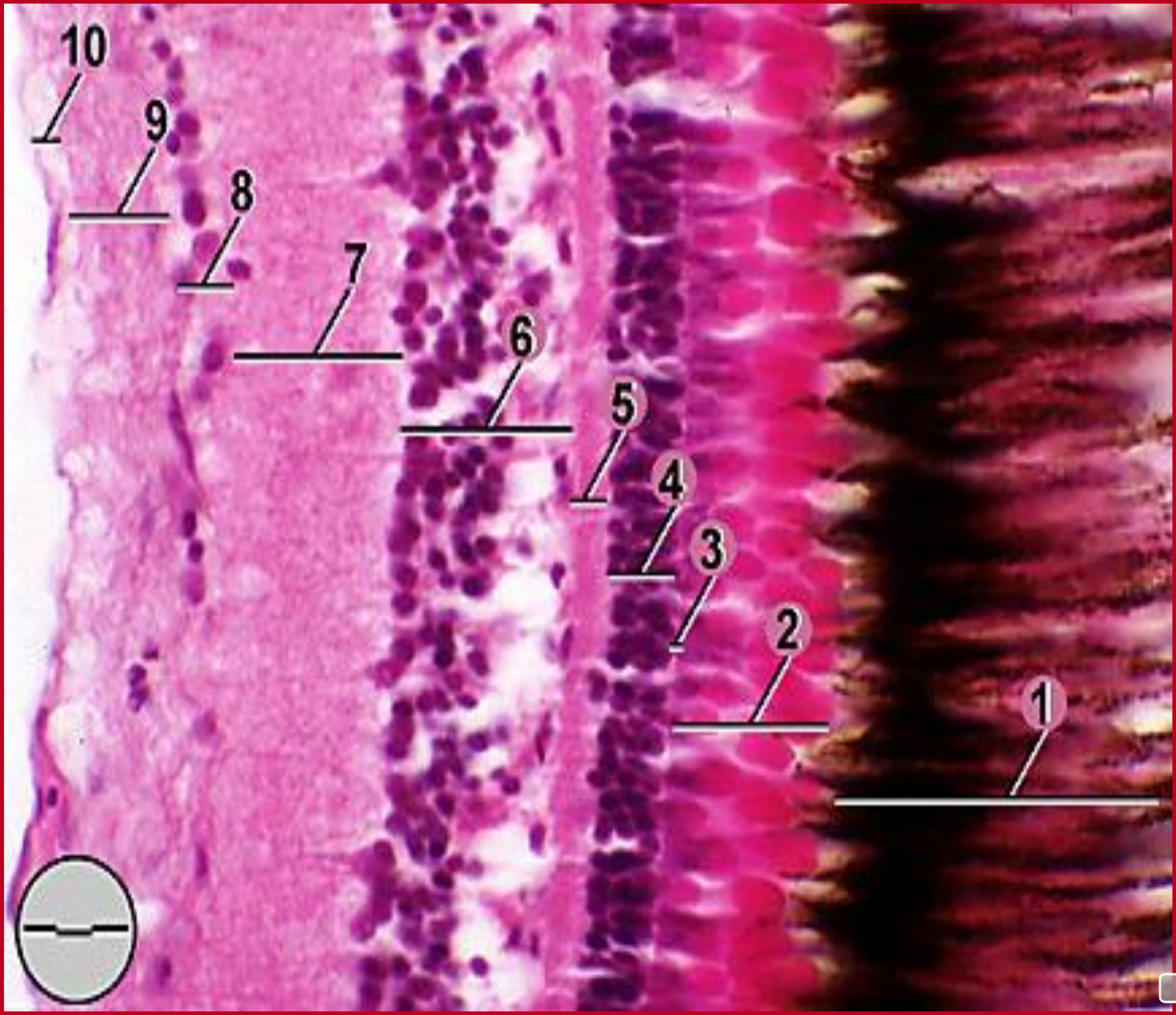
- The stroma of the iris consists of loose connective tissue rich in blood vessels and melanocytes. The amount of pigment cells in the stroma of the iris determine colour of the eye, many pigment cell: brown eye
- Some pigment cell: green eye
- Few or non : blue eye
- The posterior surface of the iris is covered by two layer of epithelium.
- The iris has 2 muscles sphincter muscle and dilator muscle which responsible for regulation the size of the pupil hence the amount of light entering the eye.



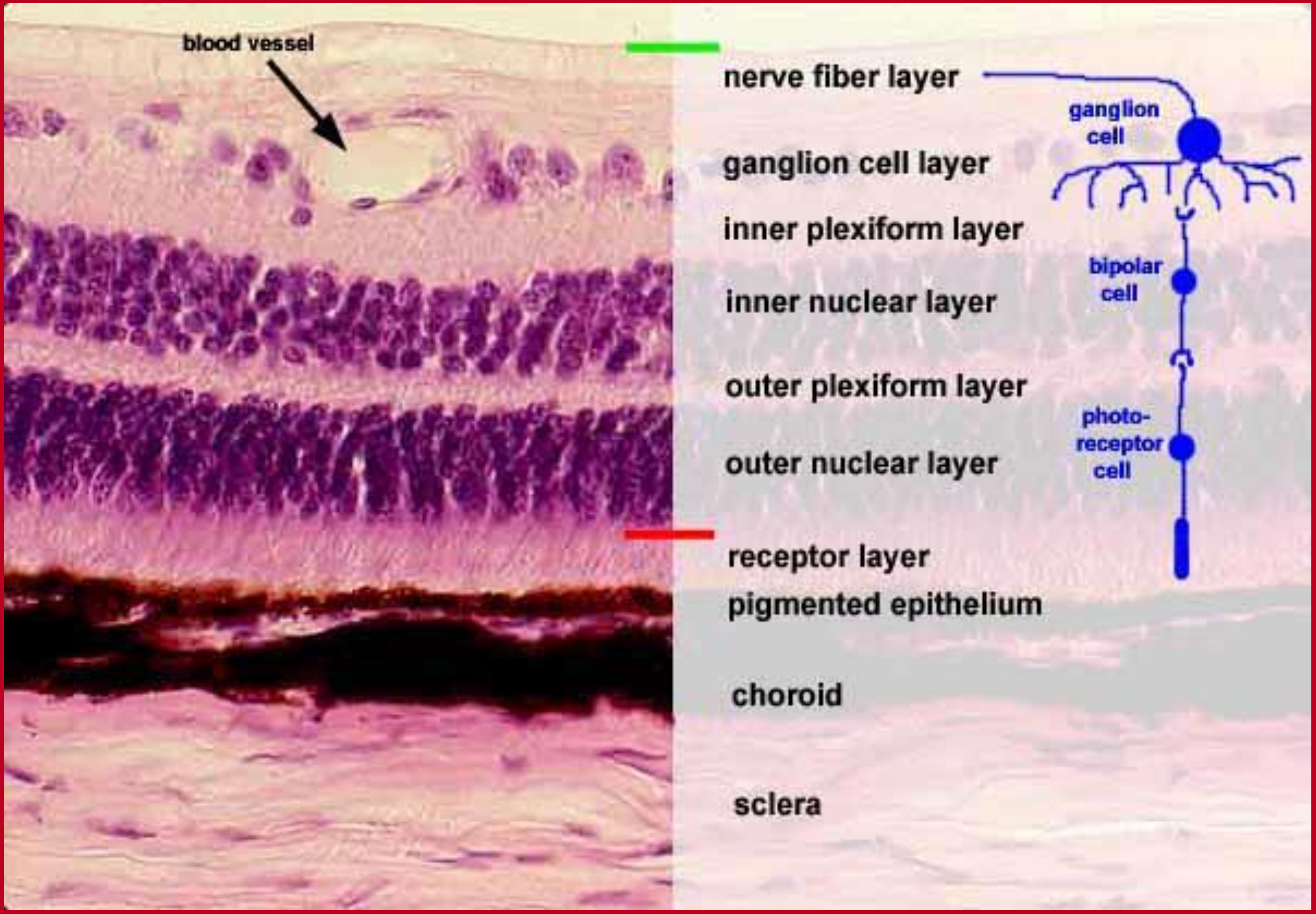
pupillary dilator
anterior chamber
stroma
pupillary sphincter
posterior chamber
pigmented epithelial layer
lens

III Tunica interna (Retina)

- The retina is the innermost layer of the eye and is the photoreceptor organ.
- The retina in transverse section shows ten layers from external to internal they are:-
- **1) Retinal pigment epithelium “pigmented retina ‘’**
- **2) layer of rods and cones "photoreceptor’’**
- **3) External limiting membrane**
- **4) The outer nuclear layer**
- **5) The outer plexiform layer**
- 6) **inner nuclear layer:**
- 7) **The inner plexiform layer**
- 8) **The ganglion cell layer**
- 9) **The optic nerve fiber layer**
- 10) **The inner limiting membrane**



10
9
8
7
6
5
4
3
2
1


blood vessel
nerve fiber layer
ganglion cell
ganglion cell layer
inner plexiform layer
bipolar cell
inner nuclear layer
outer plexiform layer
photo-receptor cell
outer nuclear layer
receptor layer
pigmented epithelium
choroid
sclera

<u>1- Retinal pigment epithelium</u> “pigmented retina”

- This melanin-rich simple cuboidal epithelium derives from optic cup.
- It rests on a thick elastic basement membrane (Bruch’s membrane) that separates it from choroids.
- This epithelium has an extensive network of tight junctions between cells which form a barrier between the blood and the neural retina.
- The dense melanin pigment absorbs light not captured by photoreceptor cell.
- Retinal pigment epithelium has a role in maintance of photoreceptor cells.

2) layer of rods and cones "photoreceptor"

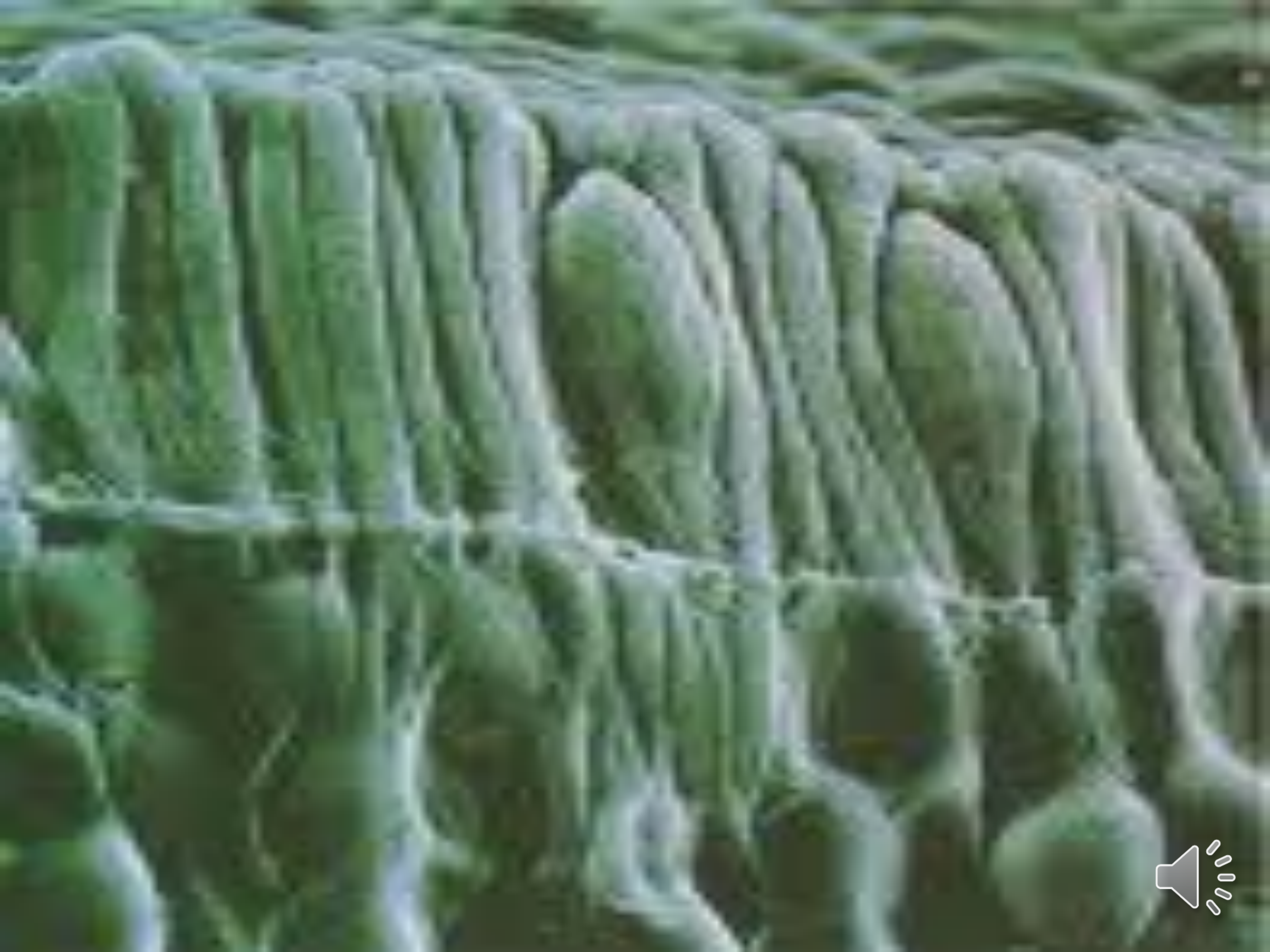
- It consists of two types of modified neurons rods and cones.
- Rods are long narrow cells, they are essential for dim-light vision/night vision.
- Cones are conical shape cells, shorter than rods and they are essential for visual acuity and color vision.
- Rods are more numerous than cones . Rods are about 120 million while cones are about 7 million.
- By electron microscope both rods and cones are consisted of:

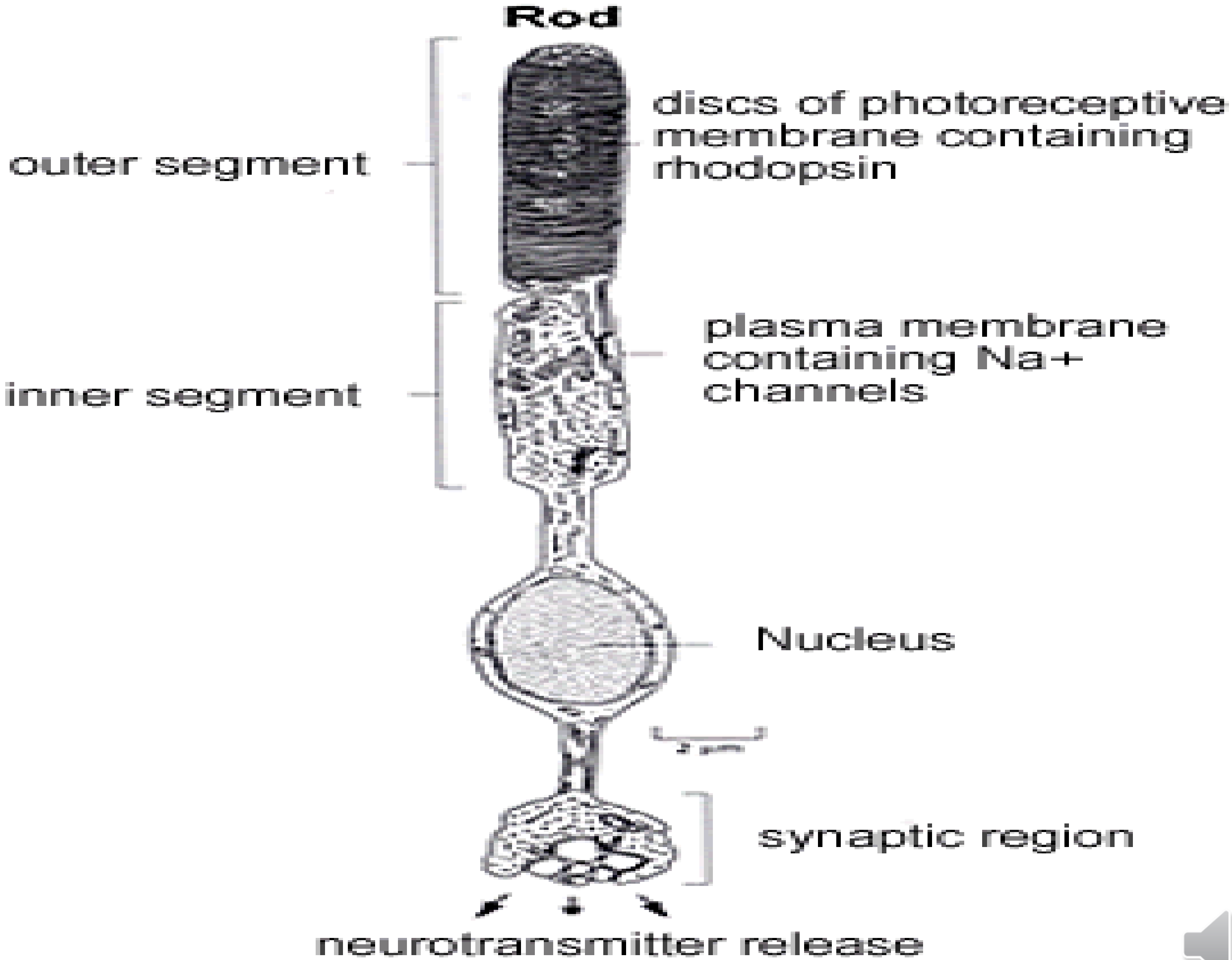
An outer segment filled with stacks of membranes containing the visual pigment molecules such as rhodopsin.

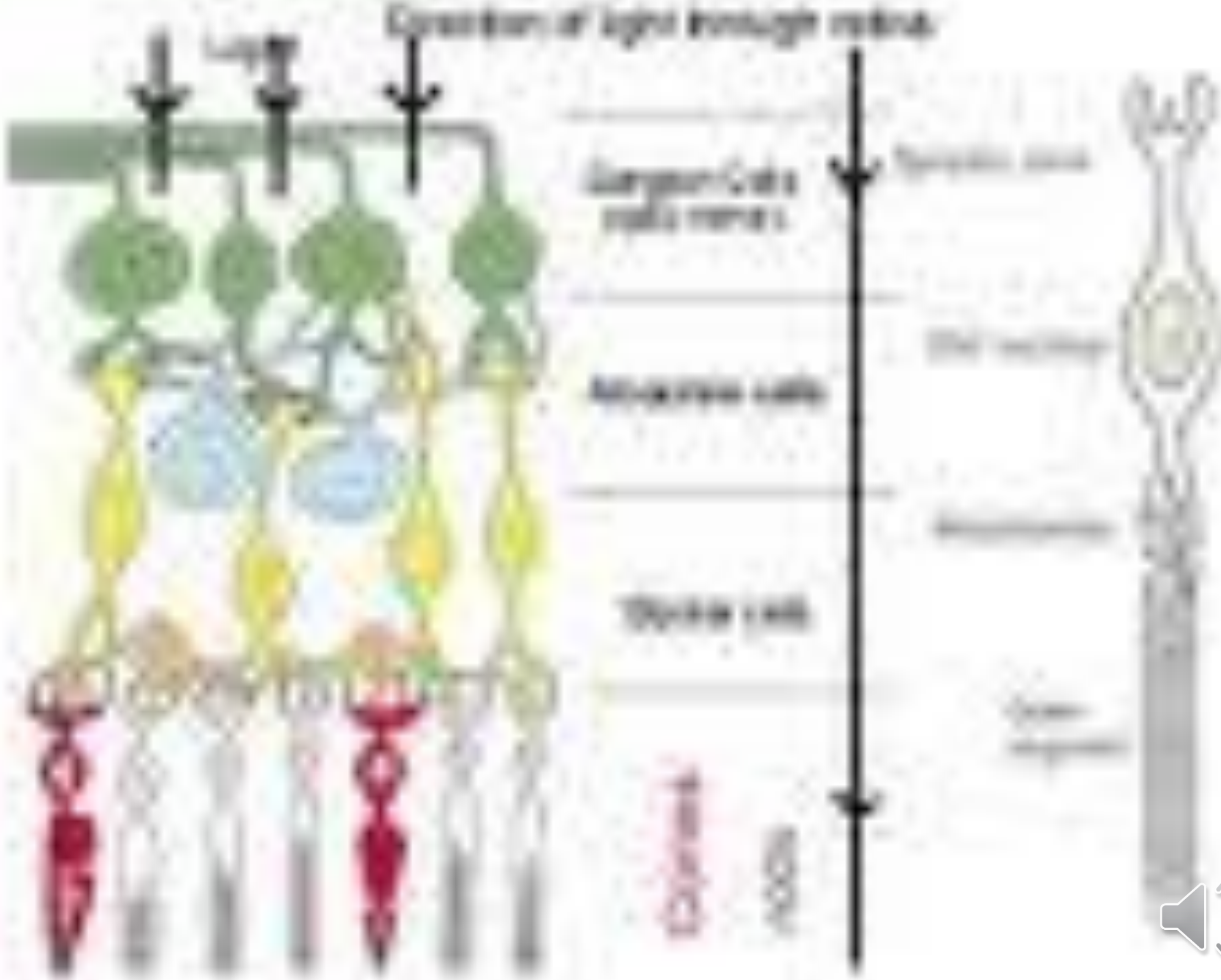
An inner segment containing numerous mitochondria in their outer part "ellipsoids" while in their inner part "myoids" contains Golgi complex, rough and smooth endoplasmic reticulum and glycogen.

A cell body containing the nucleus of the photoreceptor cells.

A synaptic terminal, where neurotransmission occurs.

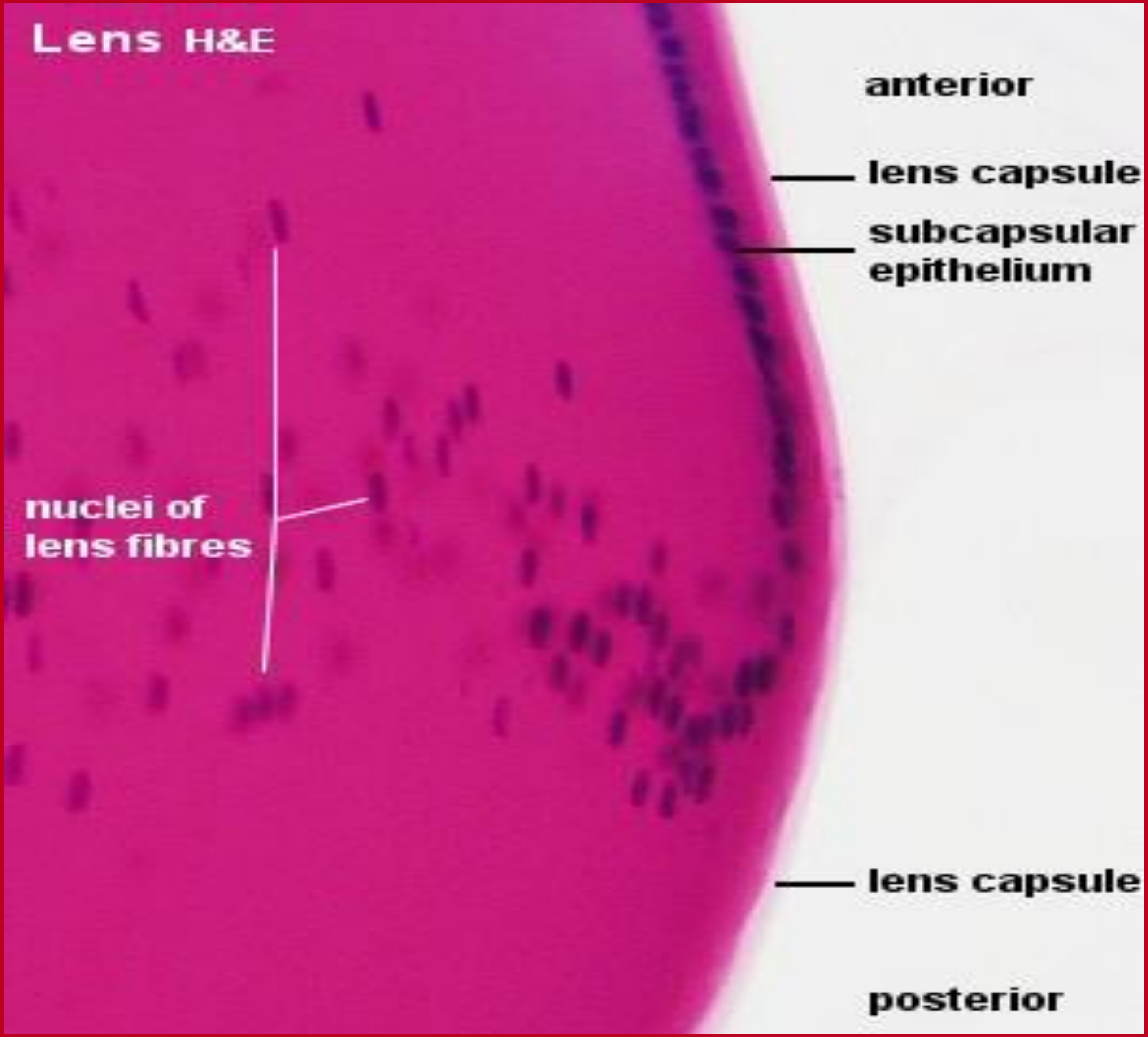



Rod
outer segment
discs of photoreceptive
membrane containing
rhodopsin
inner segment
plasma membrane
containing Na+
channels
Nucleus
synaptic region
neurotransmitter release

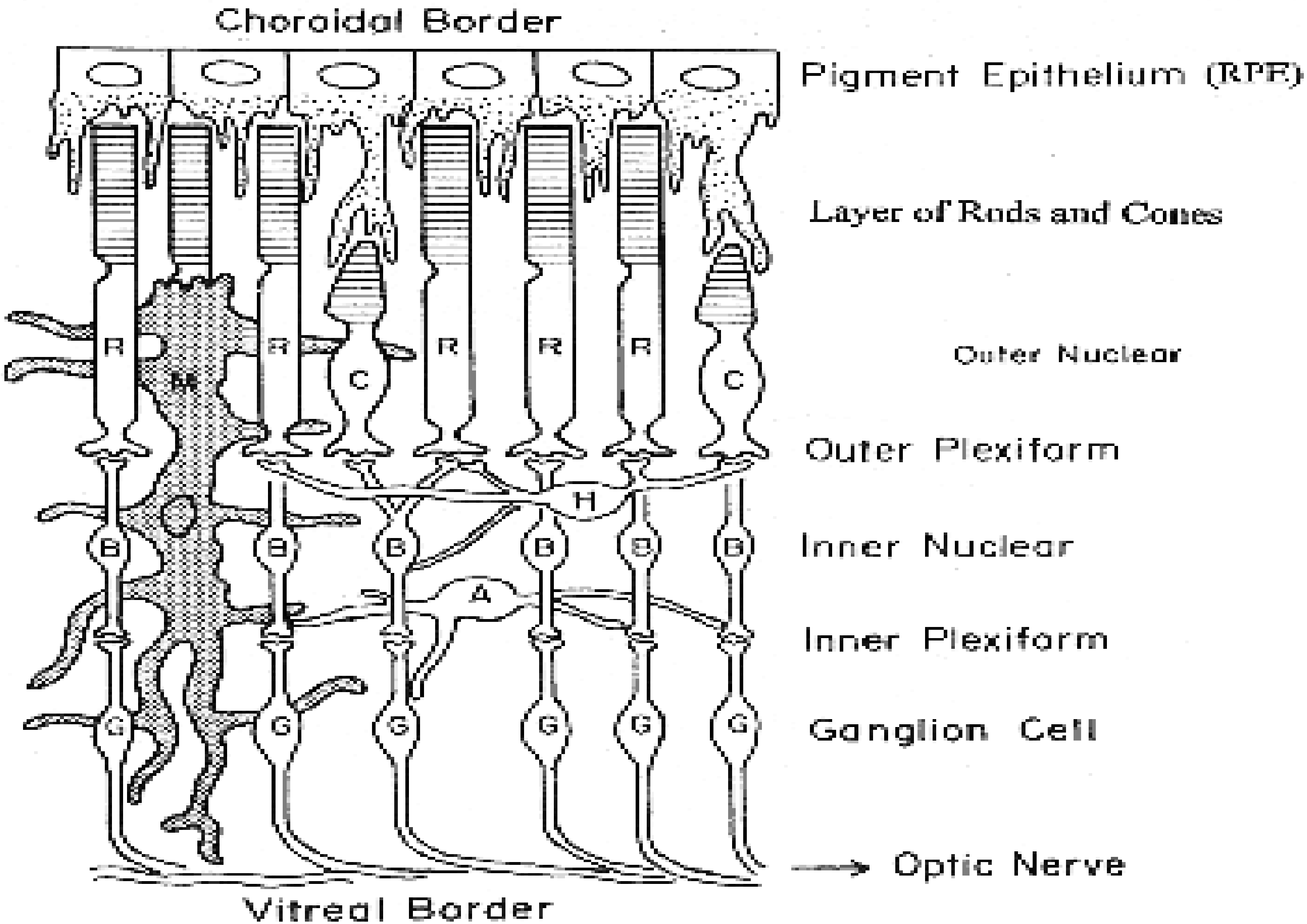

Direction of light through retina

Lens

- It has 3 main components:-
  - **<u>The lens capsule:</u>** is a very thick basement membrane (10-20 mm) that covers the entire lens. It consists of type III and type IV collagen fibrils embedded in glycoprotein.
  - **<u>Sub capsular epithelium:</u>** consists of a single layer of cuboidal epithelial cell, found only on the anterior surface of the lens. It is the site of actively dividing cells that give rise to lens fibers.
- **<u>Lens fibers:</u>** are long narrow hexagonal, specialized epithelial cells that makes up most of the lens. During differentiation, they lose their nuclei, fill with protein called crystallins and develop a variety of plasma membrane specializations



Lens H&E
anterior
lens capsule
subcapsular
epithelium
nuclei of
lens fibres
lens capsule
posterior



R Rod photoreceptor cell
C Cone photoreceptor cell
H Horizontal cell
B Bipolar cell
A Amacrine cell
G Ganglion cell
M Muller cell

Retina

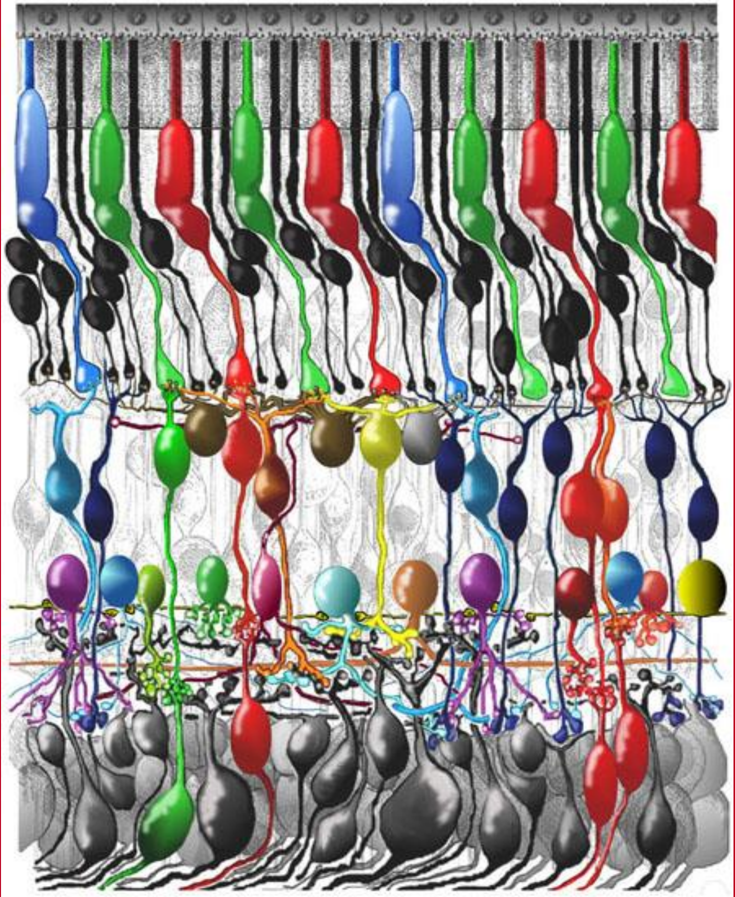


Fig. 5. Drawing of a vertical section through the human retina to show the organization of the different neurons and glial cells constituting it.

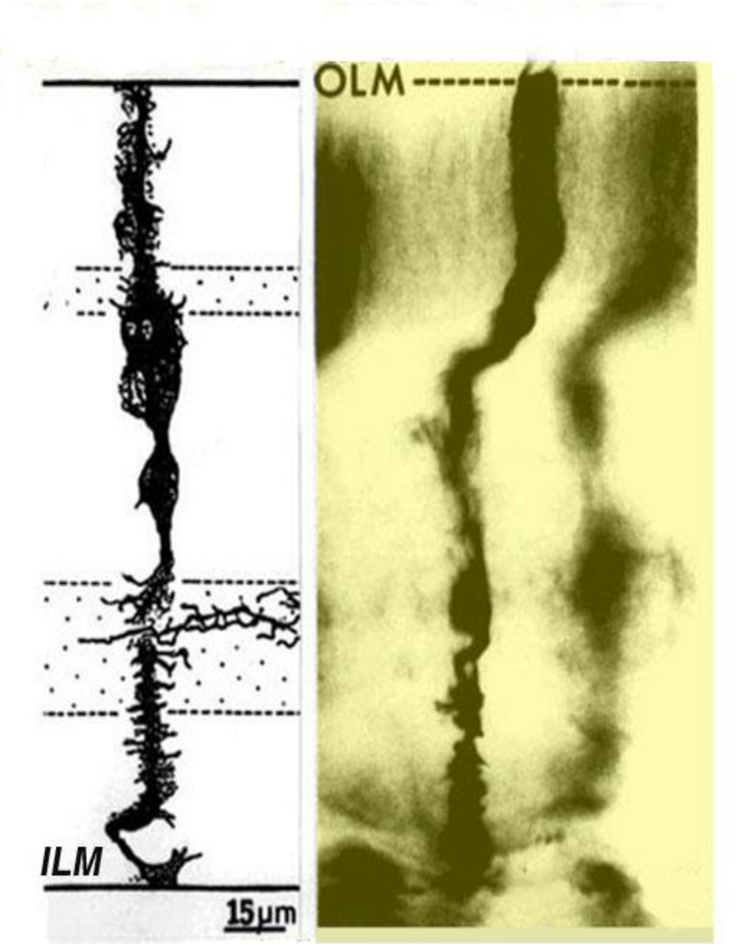


Fig. 11. Vertical view of Golgi stained Müller glial cells.

Accessory organs of the eye

Conjunctiva: This structure has 2 parts:

- The bulbar conjunctiva: is a thin nonkeratinized stratified squamous epithelium covering the eye's anterior surface to the cornea.
- The palpebral conjunctiva: is a stratified columnar epithelium covering the inner surface of the eye lids.

- The conjunctiva is under lain by a loose, vascular lamina propria containing many lymphocytes. Plasma cell and macrophage.

Eye lids :-

- It consists of anterior part and posterior part.
- The anterior part (facing the world) is also called cutaneous part, is covered by skin and contains sweat glands, sebaceous gland and along margin of the lids is 3-4 rows of hair (eyelashes).
- The posterior part (facing the eye ball) “conjunctival part” is lined by conjunctiva. Beneath the conjunctiva large sebaceous gland embedded in a plate of dense connective tissue containing many elastic fibers. The plate and sebaceous gland within it are called meibomian glands.
- **Lacrimal glands:**
- Each gland is composed of many serous acini that drain into the orbit.
- They produce tears, that clean and lubricate cornea.

End